

Final Status: α^{-1} and $\sin^2 \theta_W$

Proof levels, bridge assumptions, and remaining conditions

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1 Fine-Structure Constant $\alpha^{-1} = 137.035999177549$

Result. $\alpha_{\text{UBT}}^{-1} = 137.035999177549\dots, +0.026\sigma$ from CODATA/NIST 2022. No fitted parameter.

Overall status: [L1 cond.] (all bridge assumptions stated below).

1.1 Mathematical chain (all [L1] or [L0])

Result	Level	Source
$\Omega_\eta(1) = \frac{1}{24} - \frac{1}{8\pi}$	[L0]	Eisenstein $G_2(i)$
$12\pi\Omega_\eta = (\pi - 3)/2$	[L0]	algebra
$N_{\text{eff}} = 12 = 3 \times 4$	[L1]	su2_twist_neff12.tex
$Z_\psi[\tau] = \eta^{-12}$ (exact Gaussian)	[L1]	modular_symmetry_rpsi.tex
$R_\psi = 1$ (modular fixed point)	[L1]	modular_symmetry_rpsi.tex
$r_{\text{L2}} = 3(1 + \Omega_\eta)$	[L1]	layer2_kernel_derivation.tex
$\rho = \exp(-C_Q/(2\pi n))$ (exact Gaussian)	[L1]	gap_A_proof.tex
$n^* = 137.035999177549$ (fixed point)	[L1]	alpha_derivation_complete.tex

1.2 Physical identification $n^* = \alpha^{-1}$ ([L1 cond.]

The UBT EM action on the ψ -winding sector is

$$S_{\text{EM}}^{\text{UBT}}[A] = \frac{n}{8\pi} \int F \wedge *F - i \frac{\chi}{4\pi} \int F \wedge F,$$

giving complexified coupling $\tau_{\text{EM}}^{\text{UBT}} = \chi + in$. Matching to the standard Maxwell coupling $\tau_{\text{EM}} = \theta_{\text{EM}}/(2\pi) + i \cdot 4\pi/e^2$:

$$\frac{n}{8\pi} = \frac{1}{2e^2} \implies n = \frac{4\pi}{e^2} = \alpha^{-1}. \quad [\text{L1 cond.}]$$

[em_modular_coupling_identification.tex]

1.3 Remaining conditions for unconditional [L1]

1. **EM projection condition [L1 cond.]**: The UBT compact- ψ sector must project onto the standard unit-charge $U(1)_{\text{EM}}$ line bundle. Requires: Θ_0 vacuum orbit in the EW doublet with $Y = 1$ [`electroweak_em_projection_theorem.tex`].
2. **EW non-zero minimum [L1 cond.]**: $V(\Theta)$ must have $\mu_{\text{EW}}^2 > 0$ (SSB condition). Status: proved conditionally on the UBT-compatible potential form [`theta0_nonzero_minimum_theorem.tex`]. Open: derive $\mu_{\text{EW}}^2 > 0$ from $S[\Theta]$ dynamics.

2 Weinberg Angle $\sin^2 \theta_W(M_Z) = 0.23122$

Result. $\sin^2 \theta_W(M_Z) = 0.23122 \dots$, matching experiment $0.23122(3)$ to 5 significant figures.

Overall status: [L1 cond.] within stated bridge assumptions.

2.1 Derivation chain

Boundary condition [[L1]]: From hypercharge normalisation in UBT:

$$\sin^2 \theta_W(M_{\text{UBT}}) = \frac{3}{8}.$$

Beta coefficients [[L1 cond.], `weinberg_angle_odd_spinor_threshold_beta_theorem.tex`]: From odd-spinor threshold spectrum (paired EW doublets, Grassmann measure, 3-generation structure):

$$b_1 = \frac{33}{5}, \quad b_2 = 1, \quad b_3 = -3.$$

RG running [[STD]]:

$$\alpha_i^{-1}(M_Z) = A + b_i L, \quad L = \frac{1}{2\pi} \ln \frac{M_{\text{UBT}}}{M_Z} = 5.2555 \dots$$

Hypercharge: $\alpha_Y^{-1} = \frac{5}{3} \alpha_1^{-1}$. EM assembly:

$$\alpha_{\text{EM}}^{-1}(M_Z) = \frac{8}{3} A + 12L = 127.934 \dots \quad \text{[[STD] input]}$$

$$\implies A = \frac{3}{8} [\alpha_{\text{EM}}^{-1}(M_Z) - 12L] = 24.325 \dots$$

$$\sin^2 \theta_W(M_Z) = \frac{\alpha_2^{-1}(M_Z)}{\alpha_{\text{EM}}^{-1}(M_Z)} = \frac{A + L}{127.934} = \frac{29.581}{127.934} = 0.23122 \dots \quad \checkmark$$

2.2 Bridge assumptions and sensitivity

Bridge B5 [MC]: Physical scale $M_{\text{UBT}} \approx 2 \times 10^{16}$ GeV. This is the primary open bridge. Sensitivity: $\sin^2 \theta_W$ shifts by ± 0.003 per order of magnitude in M_{UBT} . Derivation of M_{UBT} from $S[\Theta]$: open.

Bridge B9 [L1 cond.]: $\alpha_{\text{EM}}^{-1}(M_Z) = 127.934$ from $\alpha_{\text{UBT}}^{-1} = 137.036$ plus standard QED running. [STD] input used directly.

Odd-spinor threshold [L1 cond.]: beta coefficients require paired EW doublets and Grassmann measure from odd-winding sector. `[weinberg_angle_odd_spinor_threshold_beta_th`

3 Combined Status Matrix

Quantity	Value	Level	Key open condition
α^{-1}	137.035999177549	[L1 cond.]	EW doublet vacuum (Θ_0)
$\sin^2 \theta_W(M_Z)$	0.23122	[L1 cond.] (B5)	Physical scale M_{UBT}
M_{UBT}	$\approx 2 \times 10^{16}$ GeV	[MC]	Derive from $S[\Theta]$
$\mu_{\text{EW}}^2 > 0$	(SSB sign)	[L1 cond.]	Derive from $S[\Theta]$ dynamics

What is and is not claimed.

The mathematical fixed-point chain giving $n^* = 137.036$ is [L1] (exact, no approximation). The physical identification $n^* = \alpha^{-1}$ via compact-U(1) EM normalisation is [L1 cond.]. The Weinberg angle derivation is [L1 cond.] inside stated bridge assumptions.

Neither result should be advertised as unconditionally proved. The correct statement is: “*within the UBT bridge framework, $\alpha^{-1} = 137.036$ and $\sin^2 \theta_W(M_Z) = 0.231$ follow from first principles with all derivation steps at [L1] or [L1 cond.] level.*”

The two primary remaining tasks to promote to unconditional [L1]:

1. Derive M_{UBT} (physical scale) from $S[\Theta]$ (B5).
2. Derive $\mu_{\text{EW}}^2 > 0$ (SSB sign) from $S[\Theta]$ dynamics.

References

- [1] D. Jaros, `alpha_derivation_complete.tex`, 2026.
- [2] D. Jaros, `em_modular_coupling_identification.tex`, 2026.
- [3] D. Jaros, `electroweak_em_projection_theorem.tex`, 2026.
- [4] D. Jaros, `theta0_nonzero_minimum_theorem.tex`, 2026.
- [5] D. Jaros, `weinberg_angle_odd_spinor_threshold_beta_theorem.tex`, 2026.
- [6] D. Jaros, `weinberg_angle_mz_bridge_closure.tex`, 2026.
- [7] D. Jaros, `BRIDGE_CLOSURE_MATRIX.md`, 2026.